MIL-R-398C (AR)
AMENDMENT 4
12 July 1977
SUPERSEDING
AMENDMENT 3
14 August 1973

MILITARY SPECIFICATION

RDX

This Amendment forms a part of Military Specification MIL-R-398C dated 22 August 1962, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Paragraph 1.1: After "see 6.1", add "and 6.10"
Delete "Type A" and "Type B" and substitute "Type I" and "Type II" respectively.

Paragraph 2: Delete in its entirety and substitute:

*"2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC-C-461	-	Cloth, Twill, Uniform Cotton
PPP-B-26	-	Bag, Plastic, Polyethylene
RR-S-366	-	Sieves, Standard for Testing Purposes

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1376

Source: https://assist.dla.mil -- Downloaded: 202640-16T07:19Z
Check the source to verify that this is the current version before use.

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PUBLICATIONS

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Recommended Practice for Sampling Industrial Chemicals

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA., 19103).

(Copies of Specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

PAGES 2 and 3

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply to the extent specified herein.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite, "49 CFR 100-199 (latest revision).")"

Source: https://assist.dla.mil -- Downloaded: 202610-16T07:19Z
Check the source to verify that this is the current version before use.

Paragraph 3.1: Delete "Table I" in its entirety and substitute:

TABLE I

Test	Type I	Type II	Test Paragraph
Melting point, Centrigrade (C) min.	200.0	190.0	4.3.1
Total acetone insoluble material, % maximum, max.	0.05	0.05	4.3.2
Inorganic insoluble, %, max.	0.03	0.03	4.3.3
Insoluble particles, retained on U.S. Standard No. 60, Number of particles, sieve	5	5	4.3.4
Acidity, %, max.	0.05	0.02	4.3.5

Paragraph 4.1: Delete in its entirety and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of RDX in pounds in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization."

PAGE 4

Paragraph 4.2.1: Delete in its entirety and substitute:

"4.2.1 Lot formation. A lot shall consist of a homogeneous blend of RDX from one or more batches produced by one manufacturer, in accordance with the same specification or same specification revision under one continuous set of operating conditions. Each batch shall consist of that quantity of RDX that has been subjected to the same unit chemical or physical process. The lot shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168."

*Paragraph 4.2.2: Delete in its entirety and substitute:

"4.2.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect, except where 100 percent inspection is specified."

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Bag, Cloth				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Foreign matter		0.40%	--	Visual
102	Bag pierced or torn		0.40%	5.1	Visual
103	Not tied or improperly tied		0.40%	5.1	Visual
Minor	None Defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026/10-16T07:19Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Sealed Rubber Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Bag improperly closed		0.65%	5.1	Visual/Manual
102	Bag damaged		0.65%	5.1	Balance
103	Insufficient solution		0.65%	5.1	Visual
104	Insufficient alcohol in solution		0.65%	5.1	Hydrometer
105	Net weight		0.65%	5.2	Visual
Minor	None Defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-R-398C (AR)
AMENDMENT 4

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Polyethylene Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Seam splits when manual pressure is applied along entire length of seam		0.65%	5.1	Visual/Manual
102	Bag damaged		0.65%	5.1	Visual
Minor	None Defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Tied Polyethylene Bag		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Bag improperly closed		0.65%	5.1	Visual
102	Bag damaged		0.65%	5.1	Visual
103	Insufficient Solution		0.65%	5.1	Visual
104	Insufficient alcohol in solution		0.65%	5.1	Hydrometer
105	Net weight		0.65%	5.2	Visual
Minor	None Defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-R-398C (AR)
AMENDMENT 4

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Sealed Drum				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Sealing improper		0.65%	5.2	Visual/Manual
102	Drum damaged		0.65%	5.2	Visual
103	Alcohol solution leaking from drum		0.65%	5.2	Visual
Minor					
201	Marking missing, incorrect or illegible		0.65%	5.3	Visual"

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

PAGE 5

Paragraph 4.2.3: Delete in its entirety and substitute:

"4.2.3 Testing

4.2.3.1 Sampling. The tests depicted in 4.3.1 through 4.3.6 shall be performed on a representative sample from each batch. A representative sample of approximately (12) ounces shall be selected from each batch in accordance with ASTM Procedure E300-70 for solids. The inspection of the samples shall be in accordance with MIL-STD-1235 CSP-1 Plan, Inspection Level II, AQL 4.0%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.3. If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification and code number shall be as given in Table III.

TABLE III

Melting point determination (see 3.1)	Major Defect Code No. 04001
Alternate m.p. determination (see 3.1)	Major Defect Code No. 05001
Acetone insoluble material (see 3.1)	Major Defect Code No. 06001
Inorganic insoluble material (see 3.1)	Major Defect Code No. 07001
Insoluble particles (see 3.1)	Major Defect Code No. 08001
Acidity determination (see 3.1)	Major Defect Code No. 09001
Granulation determination (see 3.2)	Major Defect Code No. 10001"

PAGE 8

*Section 5: Delete in its entirety and substitute:

"5. Preparation for Delivery

5.1 Preservation and Packaging

5.1.1 Level A. Unless otherwise specified by the procuring activity, the RDX shall be thoroughly mixed to form a slurry or cake containing not less than 10 percent by weight of a solution

Source: https://assist.dla.mil -- Downloaded: 202610-16T07:19Z
Check the source to verify that this is the currentversion before use.

made as follows: 40 percent by weight of isopropyl alcohol and 60 percent water. Not more than 50 pounds, dry weight, of the wet RDX shall be packed in a cloth bag described in 5.1.1.2. The filled cloth bags shall be placed in a rubber bag, rubberized cloth bag or two polyethylene bags, described in 5.1.1.1, placed one inside the other. Inner and outer bags shall be securely tied with the polyethylene bags individually tied, using non-metallic tape or cord. The tops of the outer bags shall be gathered and formed into a goose-neck when being tied.

5.1.1.1 The polyethylene bags shall comply with Type II, Style 1 of Specification PPP-B-26 except that closure will not be heat sealed. The bag size shall be large enough to prevent strain on the bag when it is placed in the drum and filled; the length shall be sufficient to allow gathering of the top and folding into a gooseneck when tying. The bag seams shall meet the seam strength test with an AQL of 0.65 percent at 50 percent of the breaking strength of the polyethylene film. Seams shall also be examined by separating the bag faces and applying pressure manually along the entire length of the seam. Seams which can be opened at any point in this manner are not acceptable.

5.1.1.2 The cloth bags shall be made from white cotton twill, free of size and brighteners, complying with the general requirements of Specification CCC-C-461 and the following detailed requirements:

Yarn	Carded
Yarn Ply	Warp 12/1 Fill 18/1
Weight:	7.2 $\pm$ .5 oz/sq yd (Greige) 6.2 $\pm$ .5 oz/sq yd (after bleaching)
Yarns per inch (minimum)	Warp 78 Fill 47
Breaking Strength (minimum)	Warp 170 Fill 70
Weave	3 x 1 twill

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

A suggested bag size is 19 inches wide by 29 inches in depth. The tie tape may be attached to the bag by sewing. Alternatively, double filled gray cotton duck weighing not less than 12 oz. per square yard may be used for the bag.

5.2 Packing

5.2.1 Level A. The outer bag containing a maximum of 300 pounds dry weight of crystalline RDX shall be placed in a drum complying with Department of Transportation specification 5 or 5B of the Code of Federal Regulations 49 CFR 100-199.

5.2.2 Level B. The outer bag containing a maximum of 225 pounds dry weight of crystalline RDX shall be placed in a drum complying with DOT Specification 21C.

5.3 Marking. Containers shall be marked as required by Code of Federal Regulations 49 CFR 100-199. In addition, shipments shall be marked in accordance with Standard MIL-STD-129. Each container shall be clearly labelled with the lot number and net weight of its contents. When a container holds RDX from more than one lot, all lot numbers shall be shown on the label and each of the small bags inside the container shall be clearly and permanently labelled with the lot number of its contents."

PAGE 9

Section 6: Table II: Change "A Percent, B Percent...H Percent" to "1 Percent, 2 Percent....8 Percent"

Paragraph 6.1: Change "Class A, Class B...Class H" to "Class 1 (see 6.10), Class 2.....Class 8"

Paragraph 6.3: Add: "(c) Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171."

Paragraph 6.6: Change "Type A: and "Type B" to "Type I" and "Type II" respectively.

Paragraph 6.8: After "STANAG No. 4022", Add "and ABC-Army-STD-115"

Add new paragraph:

"6.10 In accordance with the designation of classifications, paragraph 5-222 of Defense Standardization Manual 4120.3-M, the following designation changes are implemented in this amendment:

Source: https://assist.dla.mil -- Downloaded: 202610-16T07:19Z
Check the source to verify that this is the current version before use.

"Type A and B: as specified in MIL-R-398C is changed to "Type I and II" and "Class A,B,C,D,E,F,G,H" as specified in MIL-R-398C is changed to "Class 1,2,3,4,5,6,7,8". In the next division of the affected specification, these changes will be implemented."

*The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

Custodian:
ARMY-AR
NAVY-OS
AIR FORCE-70

Preparing Activity:
ARMY-AR

Reviewing Activity:
ARMY-AR
NAVY-OS
AIR FORCE-70

Project Number: 1376-0127

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would eliminate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

POSTAGE
NECESSARY
MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-01-16T07:19Z
Check the source to verify that this is the current version before use.